

Triangle Inequality Proofs & the Hinge Theorem

Geometry Worksheet · Grade 9–11

Name: _____

Date: _____

Learning Objectives

- Write two-column proofs involving triangle inequalities using given information, the reflexive property, and theorems.
- Apply the Hinge Theorem and its Converse to compare angles and sides in triangles sharing a common side.
- Construct logical proof sequences that lead from given statements to conclusions about angle or side relationships.

Problems

1. In triangle ABC, $AB = 5$ and $BC = 8$. Which side is opposite the larger angle?

$$AB = 5, BC = 8$$

2. State the reason that justifies writing $AC \cong AC$ in a two-column proof when diagonal AC is shared by two triangles.

$$AC \cong AC$$

3. In quadrilateral ABCD with diagonal AC, you are given that $AB \cong CD$ and $AC \cong AC$. If $BC < AD$, what can you conclude about angles 1 and 4 formed at vertex C?

$$BC < AD$$

4. The Hinge Theorem states that if two sides of one triangle are congruent to two sides of another triangle and the included angle of the first is larger, what is true about the third sides?

5. In two triangles sharing side PQ, with $PR = XQ$ and $PQ = PQ$, if $m\angle RPQ = 70^\circ$ and $m\angle XPQ = 50^\circ$, which is longer: RQ or XQ?

$$m\angle RPQ = 70^\circ, m\angle XPQ = 50^\circ$$



6. Write the first two statements and reasons for a two-column proof showing $m\angle 1 < m\angle 4$ in quadrilateral ABCD, given $AB \cong CD$ and $BC < AD$.

$$AB \cong CD, BC < AD$$

7. In triangle DEF, $m\angle D = 95^\circ$, $m\angle E = 50^\circ$, and $m\angle F = 35^\circ$. List the sides in order from shortest to longest.

$$m\angle D = 95^\circ, m\angle E = 50^\circ, m\angle F = 35^\circ$$

8. Given triangles ABC and DEF where $AB = DE = 7$, $AC = DF = 10$, and $m\angle A = 80^\circ$, $m\angle D = 65^\circ$, use the Hinge Theorem to compare BC and EF.

$$AB = DE = 7, AC = DF = 10, m\angle A = 80^\circ, m\angle D = 65^\circ$$

9. In quadrilateral PQRS with diagonal PR, you are given $PQ \cong SR$, $PR \cong PR$, and $QR > PS$. Write a complete four-step two-column proof to show $m\angle QPR > m\angle SRP$.

$$PQ \cong SR, QR > PS$$

10. In triangles ABD and CBD sharing side BD, with $AB = CB$ and given that $m\angle ABD > m\angle CBD$, prove that $AD > CD$ using a full two-column proof and identify all theorems used.

$$AB = CB, m\angle ABD > m\angle CBD$$



Triangle Inequality Proofs & the Hinge Theorem — Answer Key

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Answer Key

1. Answer: BC is opposite the larger angle (angle A > angle C)

- The Triangle Inequality theorem states the larger side is opposite the larger angle.
- Since $BC = 8 > AB = 5$, angle A (opposite BC) > angle C (opposite AB).

2. Answer: Reflexive Property of Congruence

- Any segment is congruent to itself.
- This is the Reflexive Property of Congruence.

3. Answer: $m\angle 1 < m\angle 4$ (by the Converse of the Hinge Theorem)

- Triangles ABC and CDA share side AC, and $AB \cong CD$ (given).
- Since the third side $BC < AD$, by the Converse of the Hinge Theorem, the included angle opposite BC is smaller, so $m\angle 1 < m\angle 4$.

4. Answer: The third side opposite the larger included angle is longer.

- The Hinge Theorem (SAS Inequality) compares third sides based on included angles.
- If the included angle is larger, the side opposite it must also be larger.

5. Answer: $RQ > XQ$

- Both triangles share side PQ, and $PR = XQ$ is given.
- By the Hinge Theorem, since $m\angle RPQ > m\angle XPQ$, the side opposite the larger angle is longer, so $RQ > XQ$.

6. Answer: Statement 1: $AB \cong CD$, Reason: Given. Statement 2: $AC \cong AC$, Reason: Reflexive Property.

- Always begin a proof by listing given information as Statement 1 with reason 'Given'.
- Identify shared sides or angles and use the Reflexive Property as the next statement.

7. Answer: $EF < DF < DE$

- The smallest side is opposite the smallest angle: side EF is opposite $\angle D$? No — EF is opposite $\angle D$ (95°), DF is opposite $\angle E$ (50°), DE is opposite $\angle F$ (35°).
- Wait — correct mapping: side opposite $\angle F$ (35°) = DE is shortest; side opposite $\angle E$ (50°) = DF; side opposite $\angle D$ (95°) = EF is longest. Order: $DE < DF < EF$.

8. Answer: $BC > EF$

- Two sides of triangle ABC equal two sides of triangle DEF ($AB=DE$, $AC=DF$).
- Since the included angle $m\angle A = 80^\circ > m\angle D = 65^\circ$, by the Hinge Theorem $BC > EF$.

9. Answer: $m\angle QPR > m\angle SRP$ by the Converse of the Hinge Theorem



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- Statement 1: $PQ \cong SR$, Reason: Given.
 - Statement 2: $PR \cong PR$, Reason: Reflexive Property.
 - Statement 3: $QR > PS$, Reason: Given.
 - Statement 4: $m\angle QPR > m\angle SRP$, Reason: Converse of the Hinge Theorem (SAS Inequality).
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10. Answer: $AD > CD$ by the Hinge Theorem

- Statement 1: $AB = CB$, Reason: Given.
 - Statement 2: $BD = BD$, Reason: Reflexive Property of Congruence.
 - Statement 3: $m\angle ABD > m\angle CBD$, Reason: Given.
 - Statement 4: $AD > CD$, Reason: Hinge Theorem — if two sides of one triangle are congruent to two sides of another and the included angle of the first is larger, then the opposite side is longer.
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